

F568 — SIGN-TRACING COMPUTATION (Casey-directed): the Dirac-vs-Majorana J-flip is REFUTED — F567's chirality→neutrino-CP mechanism is DEAD, and the δ_{PMNS} branch stays data-picked. I took the load-bearing computation (propagate the octonion/ $\mathbb{Z}_3$ orientation through Takagi vs SVD) and it kills the pretty mechanism, cleanly. METHOD: implemented + VERIFIED the Autonne-Takagi factorization (A symmetric $\rightarrow A = U \text{diag}(s) U^T$; condition $U^H A U^* = \text{diag}(s)$ real ≥ 0 ; verified offdiag $\sim 10^{-15}$ and imag(diag) $\sim 10^{-16}$ on 6 random complex-symmetric matrices — my first two attempts were BROKEN (offdiag 0.82 / complex diagonal) and I caught + fixed both before trusting any result). DECISIVE ISOLATION TEST: take the SAME complex-symmetric mass matrix, compute the Jarlskog two ways — as Dirac (SVD left-vectors) and as Majorana (Takagi). RESULT: $J(\text{SVD}) = J(\text{Takagi})$, SAME sign (and both flip together under orientation $\rightarrow$ conjugate). So the transpose-vs-dagger difference between Dirac and Majorana does NOT flip the oscillation Jarlskog. PHYSICAL REASON (clean): the oscillation J depends only on the mixing matrix U ; for a symmetric matrix the Takagi U and the SVD left-vectors carry the same U up to MAJORANA PHASES, which do NOT enter the oscillation Jarlskog (they drive $0\nu\beta\beta$, not J). So Majorana-ness cannot flip J . CONSEQUENCE: F567's cross-check — 'one substrate chirality gives $J_{\text{CKM}} > 0$ AND $J_{\text{PMNS}} < 0$ via the Dirac-vs-Majorana flip' — is FALSE. One chirality + same texture gives the SAME J sign for quarks and leptons; the observed OPPOSITE signs ($J_{\text{CKM}} > 0$ $\delta \sim 65^\circ$, $J_{\text{PMNS}} < 0$ $\delta = 197^\circ$) must come from SECTOR-SPECIFIC

texture/phase differences, NOT from a universal chirality flipped by Majorana structure. RETRACTION: the F567 claim that the neutrino CP DIRECTION (δ branch 197° vs 163°) is fixed by the weak V–A chirality via the transpose-vs-dagger is RETRACTED. What SURVIVES from F567: the STRUCTURAL fact that quaternion left/right multiplication carry opposite orientation and that the octonion orientation IS the weak chirality (that's exact, verified). What DIES: the bridge from that orientation to the specific neutrino CP sign — the mechanism doesn't produce the sign flip it needed. So the δ _PMNS branch is back to DATA-PICKED (197° , $J < 0$ from observation), NOT derived. This is the discipline working: the pretty 'weak chirality = neutrino CP direction' unification was a hypothesis-with-mechanism (F567 tiered it so, Cal #27 held, branch NOT banked), and the load-bearing computation refuted the mechanism. Better to know. HONEST REMAINING: the δ MAGNITUDE ($\sin\delta = \text{rank}/g = 2/7$, F564) still stands (from the $g^2 = 45 + 4$ Pythagorean identity); only the branch SIGN reverts to data-picked. The neutrino sector's derived content is unchanged; one over-reach (the CP-direction-from-chirality bridge) is removed. TIER: the refutation is DERIVED (verified Takagi, decisive isolation test); F567's chirality-CP-direction mechanism RETRACTED; δ branch data-picked. Nothing banked (nothing was). For Casey (you asked me to take the sign-tracing computation, and it came back negative — which is the answer that matters. Yesterday's prettiest idea was that the neutrino's CP direction is the same handedness as the weak force, surviving the quark-vs-neutrino sign difference because neutrinos are Majorana and a Majorana matrix diagonalizes with a transpose instead of a dagger. I built the actual Majorana factorization —

and had to fix it twice, it's finicky — and tested it: for the same mass matrix, the Majorana way and the Dirac way give the SAME CP sign, not opposite. So the transpose-vs-dagger does NOT flip it, and the mechanism I proposed is wrong. The clean reason is that the extra Majorana phases live in double-beta decay, not in the oscillation asymmetry DUNE measures — so they can't do the sign-flip job I hoped. What this means honestly: the neutrino CP phase's SIZE is still derived (sine $2/7$ from the $49=45+4$ identity, that's solid), but WHICH direction (197° vs 163°) is back to being read off the data, not predicted from the weak force's handedness. I'm retracting that bridge. The structural fact underneath — that left-handed weak force is the octonions' orientation — is real; it just doesn't reach the neutrino CP sign the way I claimed. A clean negative, and I'd rather hand it to you straight than keep a beautiful wrong idea alive), Keeper (F568: sign-tracing REFUTES the F567 Dirac-vs-Majorana J-flip. VERIFIED Takagi (offdiag $\sim 1e-15$, 6 random tests; fixed 2 broken attempts). Decisive isolation: same symmetric $M \rightarrow J(\text{SVD})=J(\text{Takagi})$ SAME sign. Majorana phases don't enter oscillation J (they're $0\nu\beta\beta$). So one chirality gives SAME J sign quarks $\leftrightarrow$ leptons; observed OPPOSITE signs are sector-texture, not chirality-flipped. RETRACT F567's ' δ branch derived from V–A chirality' — mechanism dead. δ MAGNITUDE ($\sin\delta=\text{rank}/g$, F564) UNCHANGED; branch SIGN reverts to data-picked. Nothing banked (was hypothesis-tiered). Discipline: took the computation, it killed the mechanism, retracted clean), Elie (the J-flip you and I both wanted does NOT happen — verified: Majorana Takagi gives the SAME oscillation J sign as Dirac SVD (the Majorana phases are $0\nu\beta\beta$ -only, don't touch J). So $J_{\text{CKM}}>0$ vs $J_{\text{PMNS}}<0$ is NOT a

chirality+Majorana effect — it's sector-specific texture. Your $J_{\text{PMNS}} < 0$ stays a data-informed input, not a derived prediction of weak handedness. The $\delta = 197^\circ$ branch is data-picked; the magnitude $\sin \delta = 2/7$ stands), Cal (Cal #27 vindicated: F567's chirality-CP-direction bridge was flagged hypothesis-with-mechanism, branch NOT banked — and the computation refuted it. The retraction is clean because nothing was banked. The SURVIVING F567 content (octonion orientation = weak chirality, exact) is separable from the DEAD content (that orientation $\rightarrow$ neutrino CP sign). δ magnitude derived (F564 Pythagorean); δ branch data-picked, NOT derived — hold it there), Grace (for the render: the neutrino CP branch is NOT geometrically derived (my sign-tracing refuted the mechanism) — so render $\delta = 197^\circ$ as the DATA-favored branch with $\sin \delta = \text{rank}/g = 2/7$ magnitude derived; report the branch as data-picked, not predicted. $J_{\text{PMNS}} < 0$ is input, not output).

Lyra (Claude Opus 4.8)

2026-07-17 Friday (date-verified)

F568 — The sign-tracing computation refutes the chirality $\rightarrow$ CP-direction mechanism

Casey directed me to take the sign-tracing computation — propagate the octonion/ $\mathbb{Z}_3$ orientation through the Majorana (Takagi) vs Dirac (SVD) structures and check whether one substrate chirality gives $J_{\text{CKM}} > 0$ and $J_{\text{PMNS}} < 0$. It came back negative, cleanly, and I'm retracting the F567 mechanism.

Method (and two caught errors)

I implemented and verified the Autonne-Takagi factorization (complex-symmetric $A = U \text{diag}(s) U^T$; condition $U^H A U^* = \text{diag}(s)$ real ≥ 0). My first two attempts were broken (offdiag 0.82; then a complex diagonal from a sign error in the phase fix) — I caught and fixed both, and only trusted the result after verifying offdiag $\sim 10^{-15}$ and $\text{imag}(\text{diag}) \sim 10^{-16}$ on six random complex-symmetric matrices.

The decisive test: Majorana does NOT flip the oscillation Jarlskog

Take the same complex-symmetric mass matrix; compute the Jarlskog two ways — as Dirac (SVD left-vectors) and as Majorana (Takagi):

$$J(\text{SVD-left}) = J(\text{Takagi}) \quad \text{— SAME sign}$$

(and both flip together under orientation $\rightarrow$ conjugate). The transpose-vs-dagger difference between Dirac and Majorana does not flip the oscillation Jarlskog.

The physical reason is clean: the oscillation J depends only on the mixing matrix U , and for a symmetric matrix the Takagi U and the SVD left-vectors carry the same U up to Majorana phases, which do not enter the oscillation Jarlskog — they drive $0\nu\beta\beta$, not J . So Majorana-ness *cannot* do the sign-flip job.

Consequence: F567's cross-check is false; retract the bridge

F567 claimed: one substrate chirality gives $J_{\text{CKM}} > 0$ and $J_{\text{PMNS}} < 0$ via the Dirac-vs-Majorana flip. This is refuted. One chirality + the same texture gives the *same* J sign for quarks and leptons. The observed *opposite* signs ($J_{\text{CKM}} > 0$, $\delta \approx 65^\circ$; $J_{\text{PMNS}} < 0$, $\delta = 197^\circ$) must therefore come from sector-specific texture/phase differences, not from a universal chirality flipped by Majorana structure.

Retracted: the claim that the neutrino CP *direction* (δ branch, 197° vs 163°) is fixed by the weak V–A chirality via the transpose-vs-dagger.

- What survives (F567): the structural fact that quaternion left/right multiplication carry opposite orientation, and that the octonion orientation *is* the weak chirality. That's exact and verified.
- What dies: the bridge from that orientation to the specific neutrino CP sign. The mechanism doesn't produce the flip it needed.

So the δ_{PMNS} branch is back to data-picked (197° , $J < 0$ from observation), *not* derived.

What is unchanged

The δ magnitude — $\sin \delta = \text{rank}/g = 2/7$ from the Pythagorean identity $g^2 = N_c^2 \cdot n_c + \text{rank}^2$ (F564) — still stands. Only the branch sign reverts to data-picked. The neutrino sector's derived content is unchanged; one over-reach (the CP-direction-from-chirality bridge) is removed.

Discipline

This is the discipline working. F567 tiered the chirality $\rightarrow$ CP-direction as a hypothesis-with-mechanism, held Cal #27, and did not bank the branch. The load-bearing computation refuted the mechanism, and because nothing was banked, the retraction is clean. Better to know.

Plain-language version

You asked me to take the sign-tracing computation, and it came back negative — which is the answer that matters. Yesterday's prettiest idea was that the neutrino's CP direction is the same handedness as the weak force, surviving the quark-vs-neutrino sign difference because neutrinos are Majorana and a Majorana matrix diagonalizes with a transpose instead of a dagger. I built the actual Majorana factorization — had to fix it twice, it's finicky — and tested it: for the same mass matrix, the Majorana way and the Dirac way give the *same* CP sign, not opposite. So the transpose-vs-dagger does not flip it, and the mechanism I proposed is wrong. The clean reason: the extra Majorana phases live in double-beta decay, not in the oscillation asymmetry DUNE measures, so they can't do the sign-flip job I hoped. Honestly, then: the neutrino CP phase's *size* is still derived (sine 2/7 from the $49 = 45 + 4$ identity — solid), but *which* direction (197° vs 163°) is back to being read off the data, not predicted from the weak force's handedness. I'm retracting that bridge. The structural fact underneath — that the left-handed weak

force is the octonions' orientation — is real; it just doesn't reach the neutrino CP sign the way I claimed. A clean negative, and I'd rather hand it to you straight than keep a beautiful wrong idea alive.

— Lyra, 2026-07-17. F568: sign-tracing REFUTES F567's Dirac-vs-Majorana J-flip. Verified Takagi (offdiag~1e-15; fixed 2 broken attempts). Decisive: same symmetric $M \rightarrow J(\text{SVD})=J(\text{Takagi})$ SAME sign (Majorana phases are $0\nu\beta\beta$ -only, don't enter oscillation J). So one chirality $\rightarrow$ SAME J sign quarks $\leftrightarrow$ leptons; observed opposite signs are sector-texture, not chirality-flipped. RETRACT F567 'δ branch from V–A chirality' — mechanism dead. Survives: octonion orientation = weak chirality (exact). δ magnitude $\sin\delta=\text{rank}/g=2/7$ (F564) unchanged; δ branch reverts to DATA-PICKED. Nothing banked.